

EMILY YU

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Address: Gorlaeus Building, Einsteinweg 55, 2333 CC Leiden, Netherlands

RESEARCH INTERESTS

Formal verification and automated reasoning, especially for hardware designs and AI-based systems;
Runtime monitoring and control: ensuring safety and correctness of complex dynamical systems.

EDUCATION

Johannes Kepler University Linz, Austria 2019 – 2023

Ph.D. in Computer Science *with distinction*

Thesis: *Certifying Hardware Model Checking*

Supervisor: Prof. Armin Biere

Imperial College London, United Kingdom 2014 – 2018

M.Eng in Computing, *First Class Honours*

Integrated Master's degree: 3-year Bachelor + 1-year Master

Thesis: *Model Checking Temporal Epistemic Logic under Bounded Recall*

Supervisor: Prof. Alessio Lomuscio

Concord College, United Kingdom 2012 – 2014

GCE A-levels (High School)

PROFESSIONAL EXPERIENCE

Leiden University, Netherlands Since Oct 2025

Tenured Assistant Professor

- Leiden Institute of Advanced Computer Science

Institute of Science and Technology Austria, Austria Sept 2023 – Sept 2025

Postdoctoral Researcher

- Working with Prof. Thomas A. Henzinger on runtime monitoring and neural control verification.

Woodpecker Technologies, Singapore Aug 2022 – Jan 2023

External Consultant

- Pre-silicon formal verification for chip designs.

Johannes Kepler University Linz, Austria Feb 2019 – Jun 2023

Project Assistant

- Formal Models and Verification Group, lead by Prof. Armin Biere.

Morgan Stanley, London, United Kingdom April – Sept 2017

Software Engineer

- Full-stack development during a 6-month industrial placement.

BlackRock, London, United Kingdom Jun – Aug 2016

Software Engineer Intern

- Worked on internal web platform development.

Imperial College London, United Kingdom

2015 – 2016

Teaching Assistant

- Part-time employment. Courses: Discrete Maths, Logic, and Reasoning about Programs.

Imperial College London, United Kingdom

Jul – Sept 2015

Research Assistant

- 20 hours per week. Group of Prof. Chris Hankin. Research on games and abstractions for cybersecurity.

CAREER BREAKS

Maternity leave 2021-2022, and 2026.

HONORS & AWARDS

- **Distinguished Paper Award**, Int. Conf. on Computer Aided Verification (CAV), 2025
- **FWF ESPRIT Fellowship**, Principal Investigator, 2025
RAVENS: Runtime Assurance and Verification of Neural Systems
Austrian Science Fund, EUR 346,505
Host Institute: Vienna University of Technology, Austria
- **Disruptive Idea Award**, Neuro-symbolic Systems (NeuS) Conference, 2025
DARPA award, USD 100,000

LANGUAGE PROFICIENCY

English (fluent), Cantonese (fluent), German (B2), Mandarin (native)

ACADEMIC SERVICES AND OUTREACH

Member of Program Committee

- AAAI Conference on Artificial Intelligence (AAAI), 2027;
- SETSS 2026 Spring School on Engineering Trustworthy Software Systems, 2026;
- Int. Conference on Computer Design (ICCD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2026;
- Int. Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026);
- Int. Symposium of Electronics Design Automation (ISED), 2026;
- Post-AI Formal Methods Workshop, 2026;
- AAAI Conference on Artificial Intelligence (AAAI), 2026;
- European Conference on Artificial Intelligence (ECAI), 2025;
- Int. Joint Conference on Artificial Intelligence (IJCAI), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2025;
- Int. Conference on integrated Formal Methods (iFM), 2024.

Member of Artifact Evaluation Committee

- Int. Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI), 2024;
- Int. Symposium on Formal Methods (FM), 2024;
- Int. Symposium on Automated Technology for Verification and Analysis (ATVA), 2024.

Journal Referee

- Science of Computer Programming, 2025.
- Journal of Artificial Intelligence Research (JAIR), 2025
- Formal Aspects of Computing Journal, 2025.

Conference (sub-)Referee

- IEEE Int. Symposium on Circuits and Systems (ISCAS) 2026;
- Int. Conference on Neuro-symbolic Systems (NeuS), 2025;
- Int. Conference on Automated Deduction (CADE), 2025.

Scientific Referent for Funding Agencies

- French National Research Agency (ANR), 2025.

Event Organization

- CAV 2021, Virtual. A member of the volunteering team.
- QONFEST 2020, Vienna, Austria. A member of the volunteering team.
- Kurt Goedel Conference 2019, Vienna, Austria. A member of the volunteering team.

Other Committees

- Committee member of the Diversity Committee of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.
- Committee member of the LIAXX (women's network) of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.

PHD/POSTDOC SUPERVISION

Harshul Gupta (PhD student) at Leiden University, starting June 2026.

Yanni Dong (Postdoc) at Leiden University, May 2026 - Present.

STUDENT SUPERVISION

Bachelor's thesis supervisor at Leiden University for Enzo Beekman, Feb 2026-Present.

Thesis title: *Certifying Constraints for Hardware Model Checking*

Master's thesis supervisor at Leiden University for Guangyu Zhu, Feb 2026-Present.

Thesis title: *Signal First-Order Logic Neural Control*

Master's thesis supervisor at Leiden University for Zhengqi Lang, Oct 2025-Present.

Thesis title: *Robust Certificates for Neural Control*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Spring 2025. Project topic: *Verification of Logic Gate Neural Networks*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Sergei Pankratov, Spring 2025. Project topic: *Certificate-based Neural Control*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Winter 2024. Project topic: *Predictive Runtime Monitoring for Complex Dynamical Systems*

TEACHING EXPERIENCE

Lecturer, **Formal Methods for AI.**

Winter 2026

Mandatory course for CS Master's students in Foundations of Computing

[This is a self-designed course by myself.](#)

Leiden University, Netherlands

Lecturer, **Software Engineering.**

Spring 2026

Leiden University, Netherlands

Teaching Assistant, **Discrete Maths, Reasoning about Programs, Logic.**

2015 - 2016

Imperial College London, London, United Kingdom.

RESEARCH VISITS & OTHERS

June-July 2024, Prof. Joost-Pieter Katoen, RWTH Aachen University, Aachen, Germany.

June-July 2024, Prof. Jürgen Giesl, RWTH Aachen University, Aachen, Germany.

Since 2023, core contributor to an Intel funded project on certifying hardware model checking, proposed by Prof. Armin Biere.

Former member of the ERC project VAMOS (2023-2025); PI: Prof. Thomas A. Henzinger.

Core developer of certificate checkers used for hardware model checking competitions.

Former member of an EPSRC project titled Games and Abstractions (2015), under the supervision of Prof. Chris Hankin at Imperial College London.

ACADEMIC REFERENCES

Prof. Thomas A. Henzinger (Institute of Science and Technology Austria) – tah@ist.ac.at

Prof. Armin Biere (University of Freiburg, Germany) – biere@cs.uni-freiburg.de

Prof. Keijo Heljanko (University of Helsinki, Finland) – keijo.heljanko@helsinki.fi

Prof. Martina Seidl (Johannes Kepler University Linz, Austria) – martina.seidl@jku.at

Note: † authorship is alphabetical.

1. Nils Froleyks, **Emily Yu**, Bart Bogaerts, Armin Biere, and Keijo Heljanko. Liveness Proofs for Hardware Model Checking. In Computer Aided Verification (CAV), 2026.
2. Nils Froleyks, and **Emily Yu**. *Hardware Model Checking Certification with Certifaiger and Cerbtora*. In International Joint Conference on Automated Reasoning (IJCAR), 2026.
3. Marek Chalupa, Thomas Henzinger, N. Ege Saraç, and **Emily Yu**†. *Quantitative Monitoring of Signal First-Order Logic*. International Symposium on Formal Methods (FM), 2026.
4. Nils Froleyks, **Emily Yu**, Armin Biere, and Keijo Heljanko. Certifying Constraints in Hardware Model Checking. International Symposium on Formal Methods (FM), 2026.
5. Thomas A. Henzinger, Konstantin Kueffner, and **Emily Yu**†. *Formal Verification of Neural Certificates Done Dynamically*. In International Conference on Runtime Verification (RV), 2025. DOI: https://doi.org/10.1007/978-3-032-05435-7_4
6. Fabian Kresse, **Emily Yu**, Christoph H. Lampert, and Thomas A. Henzinger. *Logic Gate Neural Networks are Good for Verification*. In International Conference on Neuro-symbolic Systems (NeuS), 2025. **Disruptive Idea Award**
7. Nils Froleyks, **Emily Yu**, Mathias Preiner, Armin Biere, and Keijo Heljanko. *Introducing Certificates to the Hardware Model Checking Competition*. In Computer Aided Verification (CAV), 2025. **Distinguished Paper Award**
DOI: https://doi.org/10.1007/978-3-031-98668-0_14
8. Thomas A. Henzinger, Fabian Kresse, Kaushik Mallik, **Emily Yu**, and Đorđe Žikelić†. *Predictive Monitoring of Black-Box Dynamical Systems*. Learning for Dynamics and Control Conference (L4DC), 2025.
9. **Emily Yu**, Đorđe Žikelić, and Thomas A. Henzinger. *Neural Control and Certificate Repair via Runtime Monitoring*. In proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence, AAAI, 2025. **Selected for oral presentation.**
10. Nils Froleyks, **Emily Yu**, Armin Biere, and Keijo Heljanko. *Certifying Phase Abstraction*. In International Joint Conference on Automated Reasoning (IJCAR), pp. 284-303. Cham: Springer Nature Switzerland, 2024.
DOI: https://doi.org/10.1007/978-3-031-63498-7_17
11. Nils Froleyks, **Emily Yu**, and Armin Biere. *Ternary Simulation as Abstract Interpretation (Work in Progress)*. MBMV 2024, 27. Workshop. VDE, 2024.
12. **Emily Yu**, Nils Froleyks, Armin Biere, and Keijo Heljanko. *Towards Compositional Hardware Model Checking Certification*. In Conference on Formal Methods in Computer-Aided Design (FMCAD), p. 44, 2023.
DOI: https://doi.org/10.34727/2023/isbn.978-3-85448-060-0_12
13. Nils Froleyks, **Emily Yu**, and Armin Biere. *BIG Backbones*. In Formal Methods in Computer-Aided Design (FMCAD), pp. 162-167, 2023.
DOI: https://doi.org/10.34727/2023/isbn.978-3-85448-060-0_24
14. **Emily Yu**, Nils Froleyks, Armin Biere, and Keijo Heljanko. *Stratified Certification for k-induction*. In Conference on Formal Methods in Computer-Aided Design (FMCAD), p. 11. TU Wien Academic Press, 2022.
DOI: https://doi.org/10.34727/2022/isbn.978-3-85448-053-2_11

15. Francesco Belardinelli, Alessio Lomuscio, Vadim Malvone, and **Emily Yu**[†]. *Approximating perfect recall when model checking strategic abilities: Theory and applications*. Journal of Artificial Intelligence Research 73: 897-932, 2022.
DOI: <https://doi.org/10.1613/jair.1.12539>
16. **Emily Yu**, Armin Biere, and Keijo Heljanko. *Progress in Certifying Hardware Model Checking Results*. In Computer Aided Verification: 33rd International Conference, CAV 2021, July 20–23, 2021, Proceedings, Part II 33, pp. 363-386. Springer International Publishing, 2021.
DOI: https://doi.org/10.1007/978-3-030-81688-9_17
17. Francesco Belardinelli, Alessio Lomuscio, and **Emily Yu**[†]. *Model Checking Temporal Epistemic Logic under Bounded Recall*. In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, no. 05, pp. 7071-7078. 2020.
DOI: <https://doi.org/10.1609/aaai.v34i05.6193>
18. **Emily Yu**, Martina Seidl, and Armin Biere. *A Framework for Model Checking against CTLK using Quantified Boolean Formulas*. In Formal Techniques for Safety-Critical Systems (FTSCS), Revised Selected Papers 7, pp. 127-132. Springer International Publishing, 2020.
DOI: https://doi.org/10.1007/978-3-030-46902-3_8
19. **Zhengqi Yu**, Armin Biere, and Keijo Heljanko. *Certifying Hardware Model Checking Results*. International Conference on Formal Engineering Methods (ICFEM). Cham: Springer International Publishing, 2019.
DOI: https://doi.org/10.1007/978-3-030-32409-4_32

NON-REFEREED / NON-PROCEEDINGS

1. Thomas A. Henzinger, Kaushik Mallik, **Emily Yu**, and Đorđe Žikelić[†]. *Control Barrier Functions with Lookahead*. Submitted, 2026.
2. Fabian Kresse, **Emily Yu**, and Christoph H. Lampert. *Scalable Interconnect Learning in Boolean Networks*. arXiv preprint arXiv:2507.02585 (2025).
3. Nils Froleyks, **Emily Yu**, and Armin Biere. *Challenging Certificates from Model Checking*. SAT Competition 2025.
4. **Emily Yu**, Armin Biere, and Mathias Fleury. *Hardware Model Checking Certificates*. SAT Competition 2022.
5. Nils Froleyks, **Emily Yu**, and Armin Biere. *Unique Reconfiguration Sequence*. SAT Competition 2022.
6. Nils Froleyks, **Emily Yu**, and Armin Biere. *ReconfAIGERation entering Core Challenge*, 2022.

DISSERTATIONS

1. **Emily Yu**. *Certifying Hardware Model Checking*. PhD Dissertation. Johannes Kepler University. August 2023.

Ensuring the correctness of safety-critical systems is a pressing challenge, and formal verification techniques are becoming increasingly vital to achieve this goal. In automated reasoning, verifiers not only provide a yes/no answer but also generate certificates, which are machine-checkable proofs that can be independently validated by formally verified tools or even subjected to an additional layer of checking by trusted theorem provers. This thesis focuses on advancing these ideas in the domain of hardware model checking in order to increase trust in verification results.

INVITED TALKS/LECTURES

Dutch Formal Methods Day 2026 Keynote Speaker, <i>Automated Verification with Certificates</i>	June 2026
International School on Engineering Trustworthy Software Systems (SETSS) Chongqin, China Invited Lecturer (series of 3 lectures), <i>Certification for Trustworthy Formal Methods</i> Other invited lecturers include Moshe Y. Vardi, and Turing Award winner Joseph Sifakis, etc.	May 2026
Leiden University , Leiden, Netherlands LIACS Research Seminar <i>Symbolic Verification and Neural Control via Certification</i>	December 2025
IEEE Women in Circuits and Systems – Young Professionals Symposium Leiden, Netherlands Invited speaker and Panelist, <i>Hardware Verification for Trustworthy Circuits</i>	Nov 2025
TU Delft , Delft, Netherlands Delft Center for Systems and Control (DCSC) Colloquium <i>Symbolic Verification and Neural Control via Certification</i>	Nov 2025
Leiden University , Leiden, Netherlands <i>Trustworthy Formal Methods for Hardware Design Correctness</i>	April 2025
Lund University , Lund, Sweden <i>Trustworthy Formal Methods for Hardware Design Correctness</i>	April 2025
ETH Zürich , Zürich, Switzerland <i>Trustworthy Formal Methods for Hardware Design Correctness</i>	Mar 2025
RWTH Aachen University , Aachen, Germany <i>Certifying Phase Abstraction</i>	July 2024
ETH Zürich , Zürich, Switzerland <i>Model Checking Multi-Agent Systems under Recalls</i>	Jan 2018

OTHER PRESENTATIONS

TCS Seminar , Institute of Science and Technology Austria, Austria <i>Towards Compositional Hardware Model Checking Certification</i>	November 2023
LogiCS Seminar , TU Wien, Austria <i>Towards Compositional Certification for Hardware Model Checking</i>	May 2023
Int. Conference on Formal Methods in Computer-Aided Design , Trento, Italy <i>Stratified Certification for k-induction</i>	October 2022
Alpine Verification Meeting , Frauenchiemsee, Germany <i>Certifying Model Checking Results</i>	September 2022
Int. Conference on Computer-Aided Verification (CAV) , Virtual <i>Progress in Certifying Hardware Model Checking Results</i>	July 2021
LogiCS Seminar , TU Wien, Austria <i>Increasing Confidence in Model Checking</i>	May 2020
LogiCS Seminar , TU Wien, Austria <i>Model Checking Temporal Epistemic Logic Under Bounded Recall</i>	January 2020
Int. Conference on Formal Engineering Methods , Shenzhen, China <i>Certifying Hardware Model Checking Results</i>	November 2019

FTSCS Workshop, Shenzhen, China

November 2019

A Framework for Model Checking against CTLK using Quantified Boolean Formulas

Alpine Verification Meeting, Brno, Czech Republic

September 2019

Encoding Model Checking Problems for Multi-agent Systems

IFIP WG 2.2, Vienna, Austria

September 2019

Model Checking Temporal Epistemic Logic

INVITED WORKSHOPS

BIRS workshop

Jan 2026

Theory and Practice of SAT and Combinatorial Solving